AN ELLIPTIC SQUARE FUNCTION ESTIMATE IN THREE DIMENSIONS

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Abstract. We deduce an L^3 reverse square function estimate for positively curved surfaces in three dimensions from the discrete packet-density and Nikodym estimates stated in recent OpenAI manuscripts. We state the external results separately and prove the passage from finite arrays to arbitrary functions supported near the surface. The proof includes the construction of the frequency partition, continuous reconstruction from terminal samples, changes of packet windows, and spatial localization.

1. Introduction

Let U be a bounded open subset of $\mathbb{R}^2$, and let ϕ be a smooth real-valued function on a neighbourhood of $\overline{U}$. Suppose that

$$c|v|^2 \le v^T D^2\phi(\xi)v \le C|v|^2 \qquad (\xi \in U,\ v \in \mathbb{R}^2) \tag{1.1}$$

for positive constants c, C. Here $D^2\phi$ is the matrix of second derivatives. After decreasing the neighbourhood if necessary, we impose the same bounds there, with possibly different constants. We consider the graph

$$\Sigma = \{(\xi, \phi(\xi)) : \xi \in \overline{U}\} \subset \mathbb{R}^3.$$

All graph pieces used below are compact and admit this fixed smooth extension.

For $R \ge 2$ put $r = R^{1/2}$ and $\delta = r^{-1}$. Fix positive constants c_0, C_0. A regular family of caps consists of the sets

$$\theta_\nu = \{(\xi, \lambda) : |\xi - \nu| \le C_0\delta,\ |\lambda - \phi(\xi)| \le C_0\delta^2\}, \tag{1.2}$$

restricted to a fixed bounded extension of the parameter region, where the centres are $c_0\delta$-separated. Caps used to cover the graph are chosen so that their base disks cover $\overline{U}$. Section 2 explains both this covering and a smooth partition subordinate to it. The frequency dimensions of a cap are δ, δ, δ^2 in its tangent and normal directions.

Let f_θ be Schwartz functions on $\mathbb{R}^3$ with $\operatorname{supp} \widehat{f_\theta} \subset \theta$. A Schwartz function is smooth and, together with all its derivatives, decreases faster than every inverse power of $1 + |y|$. Set

$$f = \sum_\theta f_\theta, \qquad \mathcal{S}(\mathbf{f}) = \left(\sum_\theta |f_\theta|^2\right)^{1/2}, \qquad \mathbf{f} = (f_\theta)_\theta.$$

The square function refers to this given family; the summands need not be produced by one prescribed partition of unity.

Theorem 1. *Assume the external results stated in Section 3. For every $\varepsilon > 0$,*

$$\left\| \sum_\theta f_\theta \right\|_{L^3(\mathbb{R}^3)} \le C_\varepsilon R^\varepsilon \left\| \mathcal{S}(\mathbf{f}) \right\|_{L^3(\mathbb{R}^3)}. \tag{1.3}$$

Date: October 8, 2026.

2020 *Mathematics Subject Classification.* Primary 42B10; Secondary 42B25, 42B37.

Key words and phrases. Reverse square function, elliptic surfaces, Fourier restriction, wave packets, Nikodym maximal function.

The constant is independent of R, of the regular cap family, and of its functions. It may depend on the fixed surface, its extension, c_0, and C_0. The same conclusion holds for a compact smooth embedded surface in $\mathbb{R}^3$ whose second fundamental form is definite, including a surface with smooth boundary, with caps as defined in Section 8.

The assertion proved here is the deduction of (1.3) from the specified external results. We do not prove the propagation or Nikodym theorems themselves. In particular, no continuous packet estimate is left as an additional hypothesis: the passage from the cited discrete estimate to the continuous estimate is proved in Section 4.

The distinction between a square function and decoupling is relevant. Minkowski's inequality gives

$$\left\| \left(\sum_{\theta} |f_\theta|^2 \right)^{1/2} \right\|_{L^3} \le \left(\sum_{\theta} \|f_\theta\|_{L^3}^2 \right)^{1/2}.$$

An estimate with the latter, larger quantity does not yield (1.3). Likewise, a positive density of wave-packet coefficients need not agree pointwise with the prescribed square function: separating packets can remove cancellation within one cap. We preserve that cancellation by controlling the coefficient density through averages of $|f_\theta|^2$ and then removing the averages by duality.

The spherical reverse Littlewood–Paley problem at exponent $2n/(n-1)$ is discussed by Carbery [1]; at $n = 3$ this exponent is 3. His scale-independent formulation is stronger than the conclusion here. Guth, Wang, and Zhang [3] proved the corresponding critical L^4 estimate for the cone in $\mathbb{R}^3$, whose curvature has rank one. Our surface has two nonzero principal curvatures. The packet-density estimate used below is equation (4.10) of [5]. Its capacity decomposition, Lemma 4.2 of that paper, is part of the borrowed input, and we claim no new decomposition of that kind.

2. Frequency partitions and elementary notation

We use Lebesgue measure and the Fourier transform

$$\widehat{g}(\zeta) = \int_{\mathbb{R}^d} g(y) e^{2\pi i(-y\cdot\zeta)}\, dy, \qquad \check{g}(y) = \int_{\mathbb{R}^d} g(\zeta) e^{2\pi i(y\cdot\zeta)}\, d\zeta.$$

The dimension d is specified by the variables. We write $\|g\|_p = (\int |g|^p)^{1/p}$, $(g * h)(y) = \int g(y-z)h(z)\, dz$, and $\langle u \rangle = 1 + |u|$. The symbol $A \lesssim B$ means $A \le CB$, with C depending only on fixed geometric data and on explicitly fixed decay orders. The notation $A \asymp B$ means both inequalities. The notation C_c^∞ means smooth functions of compact support. A multi-index is a tuple of nonnegative integers specifying a partial derivative.

2.1. **A covering and its smooth partition.** Choose a maximal δ-separated set $\{\nu\}$ in $\overline{U}$. Maximality means that no further point of $\overline{U}$ can be added while maintaining the separation. Consequently the disks $B(\nu, \delta)$ cover $\overline{U}$. The disks $B(\nu, \delta/3)$ are disjoint, so comparison of areas shows that every disk of radius $A\delta$ contains at most C_A centres. Thus any fixed enlargement of the covering has bounded overlap, independently of R. There are $O(\delta^{-2}) = O(R)$ centres.

Let $\beta \in C_c^\infty(\mathbb{R}^2)$ be nonnegative, equal to one on $B(0,1)$, and supported in $B(0,2)$. On a neighbourhood of $\overline{U}$ define

$$B_\delta(\xi) = \sum_{\nu} \beta((\xi - \nu)/\delta), \qquad \psi_\nu(\xi) = \frac{\beta((\xi-\nu)/\delta)}{B_\delta(\xi)}.$$

The denominator is at least one on $\overline{U}$. The functions are smooth there,

$$\sum_{\nu} \psi_\nu = 1, \qquad \operatorname{supp} \psi_\nu \subset B(\nu, 2\delta), \qquad |\partial^\alpha \psi_\nu| \le C_\alpha \delta^{-|\alpha|}.$$

They may be multiplied by a fixed cutoff equal to one on the parameter region. Choose also $\rho_0 \in C_c^\infty(\mathbb{R})$ equal to one on the allowed interval of normal offsets. The multipliers

$$m_\nu(\xi,\lambda) = \psi_\nu(\xi)\rho_0(R(\lambda-\phi(\xi)))$$

sum to one on the thickened graph in question. Hence $\widehat{f_\nu} = m_\nu \widehat{f}$ gives one admissible decomposition. The overlapping caps are a covering of frequency space; the identity $\sum m_\nu = 1$ is the partition of unity. We do not assert that the caps are disjoint.

The theorem also allows other separated nets and arbitrary functions supported in their caps. The proof uses only bounded overlap, the number of centres, and the stated dimensions, and therefore applies to all these families with the same fixed separation constants.

2.2. **Why the normal scale is δ^2.** Write $V_\nu = \nabla\phi(\nu)$. Taylor's formula gives

$$\phi(\xi) = \phi(\nu) + V_\nu \cdot (\xi-\nu) + O(|\xi-\nu|^2).$$

On (1.2), $|\xi-\nu| = O(\delta)$ and therefore

$$|\lambda - \phi(\nu) - V_\nu\cdot(\xi-\nu)| = O(\delta^2). \tag{2.1}$$

Conversely a sufficiently small box defined by these two inequalities lies in a fixed enlargement of (1.2). Boundedness of V_ν makes these sheared boxes comparable with boxes measured in orthonormal tangent and normal coordinates. All these comparisons have constants independent of R.

Writing physical coordinates as $(x,t) \in \mathbb{R}^2 \times \mathbb{R}$, the reciprocal scales are

$$|x + tV_\nu| \lesssim r, \qquad |t| \lesssim R.$$

Such a set has volume comparable to $r^2R = R^2$. Its long direction is $(-V_\nu, 1)$, which is normal to the graph. For example, $\phi(\xi) = |\xi|^2/2$ gives elliptic paraboloid caps. A small spherical cap, after a rotation and translation, has $\phi(\xi) = -\sqrt{1-|\xi|^2}$ and satisfies (1.1) on a fixed disk strictly inside $B(0,1)$.

2.3. **Decreasing weights.** For $b > a > 2$ with $b > a+2$,

$$\sum_{n\in\mathbb{Z}^2} \langle u-n\rangle^{-b}\langle n-v\rangle^{-a} \lesssim \langle u-v\rangle^{-a}. \tag{2.2}$$

Indeed, at least one of $|u-n|, |n-v|$ is at least $|u-v|/2$. On the second region sum the first, integrable weight. On the first region remove a factor $\langle u-v\rangle^{-a}$ from the first weight and sum the remaining integrable power. The constants are uniform for fixed rectangular lattices and all their translates. The same proof works in dimension d with $a > d$ and $b > a+d$, and separately in the spatial and temporal factors of a product weight.

3. Preliminaries: the results borrowed from OpenAI

We state the external inputs in the normalizations used in the proof. Their proofs are not part of this article. No conclusion below uses a set-dimension theorem in place of a maximal operator estimate.

3.1. **The permitted phases.** Put

$$q_s(\xi) = -\sqrt{1-|\xi|^2}, \qquad Q_s(\xi) = -1 + |\xi|^2/2.$$

A continued phase is a function on $\mathbb{R}^2$ of the form

$$\Phi = Q_s + \zeta(q_s - Q_s), \quad 0 \le \zeta \le 1, \quad \zeta = 1 \text{ on } B(0,\varrho), \quad \operatorname{supp}\zeta \subset B(0,2\varrho), \tag{3.1}$$

where $2\varrho < 1$, the product is extended by zero, and Φ has uniformly positive definite bounded Hessian and bounded derivatives of every order at least two. The cutoff need not be radial.

External input 1 (Placement of elliptic graphs, [5, Lemma 3.1])**.** Every smooth graph with positive definite Hessian, in a sufficiently small neighbourhood of any one of its points, is carried by a fixed invertible affine change of ambient frequency coordinates to a graph agreeing with a continued phase (3.1). The neighbourhood, change of coordinates, and phase are independent of R and of all small loss parameters.

We first work with such a phase and write it as ϕ. Arbitrary graph pieces will be recovered by finitely many fixed coordinate changes in Section 8.

3.2. **A discrete packet-density estimate.** Fix a nonnegative $\chi \in C_c^\infty(\mathbb{R}^2)$ satisfying

$$\sum_{m\in\mathbb{Z}^2} \chi(\xi - m)^2 = 1. \tag{3.2}$$

For example, divide a bump positive on $[-1,1]^2$ by the square root of the sum of the squares of its integer translates. Choose an integer $D \geq 1$ so large that $\operatorname{supp}\chi$ is contained in the interior of a square of side D. For $a > 0$ write $\chi_\nu^a(\xi) = \chi((\xi-\nu)/a)$, where $\nu \in a\mathbb{Z}^2$.

Let $N \geq 2$ be a power of two, put $R = N^2$, $\delta = N^{-1}$, and let $d = (d_{\nu,v})$ be a finite complex array with

$$\nu \in \delta\mathbb{Z}^2, \qquad v \in (N/D)\mathbb{Z}^2.$$

Its frequency labels lie in a fixed bounded set, and initially $|v| \leq C_1 R$, with C_1 fixed. Define

$$g_d(\xi) = (D\delta)^{-2} \sum_{\nu,v} d_{\nu,v}\chi_\nu^\delta(\xi) e^{2\pi i(-v\cdot\xi)}, \tag{3.3}$$

$$F_d(x,t) = \int g_d(\xi) e^{2\pi i(x\cdot\xi + t\phi(\xi))}\, d\xi, \tag{3.4}$$

$$H_d^0(x,t) = \sum_{\nu,v} |d_{\nu,v}|^2 \mathbf{1}_{\{|x-v+t\nabla\phi(\nu)|\leq N\}}, \qquad 0 \leq t \leq R. \tag{3.5}$$

Here $\mathbf{1}_E$ denotes the indicator of a set E. The terminal coefficients are

$$c_{\ell,w,j} = \int g_d(\xi)\chi_\ell^1(\xi) e^{2\pi i(w\cdot\xi + j\phi(\xi))}\, d\xi, \quad \ell \in \mathbb{Z}^2, \quad w \in D^{-1}\mathbb{Z}^2, \quad 0 \leq j < R. \tag{3.6}$$

Only finitely many ℓ have a nonzero coefficient. Restrict w to $|w| \leq C_2 R$, where C_2 is any fixed constant chosen in advance.

External input 2 (Discrete density estimate, [5, equation (4.10)])**.** For each fixed setup above, $0 < \kappa < 1/10$, and $0 < \eta \leq 1$, there is a finite $P(\kappa,\eta) \geq 1$ such that

$$\sum_{\ell,w,j} |c_{\ell,w,j}|^3 \leq P(\kappa,\eta) N^{10\kappa+\eta} \log^2(2+N) \int_0^R \int_{\mathbb{R}^2} (H_d^0)^{3/2}\, dx\, dt. \tag{3.7}$$

The sum can be restricted to any subset of the retained indices. The bound is uniform over finite arrays and their coefficient sizes.

For comparison with the source, its position variable is $z = v/N^2$ and its spacetime variables are $(X,s) = (x,t)/N^2$. Thus $dX\, ds = N^{-6}\, dx\, dt$. Multiplication of the source's equation (4.10) by N^6 gives exactly (3.7). Its Theorem 3.4 specializes the propagation theorem of [4, Theorem 6.4]; its Lemma 4.2 supplies the capacity decomposition. We use the density estimate before the subsequent weighted Kakeya bound is inserted. All label regions in (3.7) remain fixed. Any finitely many scales below the source's fixed threshold are included by enlarging a parameter-independent constant: the index sets are then bounded, and the lower bound (4.8) controls their coefficient norms.

3.3. **The Nikodym maximal estimate.** A tube is the neighbourhood of a line segment in $\mathbb{R}^3$. If its segment has length L and its radius is τL, call its dimensions $(\tau L, \tau L, L)$. Let $\mathcal{N}_{\tau,L}h(y)$ be the supremum of the averages of $|h|$ over all such tubes containing y.

External input 3 (Critical Nikodym estimate, [6, Corollary 11.1, $p = q = 3$])**.** For every $0 < \gamma \le 1$ there is a finite $M(\gamma) \ge 1$ such that

$$\|\mathcal{N}_{\tau,1}h\|_{L^3(\mathbb{R}^3)} \le M(\gamma)\tau^{-\gamma}\|h\|_{L^3(\mathbb{R}^3)}, \qquad 0 < \tau < 1. \tag{3.8}$$

The source takes segments passing through y. If y merely lies in the tube, translating the axis by at most its radius gives a tube through y containing the original tube after a fixed enlargement. Thus the two conventions are equivalent up to constants. Dilation gives the same bound for every fixed $L > 0$. Fixed enlargements can be absorbed into $M(\gamma)$ uniformly for $0 < \gamma \le 1$. We do not take a supremum over L inside (3.8).

Neither input states a polynomial bound for $P(\kappa, \eta)$ or $M(\gamma)$ as its parameters approach zero. We retain these functions in the estimates below.

4. From discrete samples to a continuous packet estimate

Throughout this section $R = N^2$, with N a power of two. Fix an integer $M_0 > 10$ and, in the weights below, a fixed exponent $M \ge M_0$. The letter M without an argument is this decay order; $M(\gamma)$ always denotes the external constant in (3.8).

For a fixed smooth function b of compact support define

$$P^b_{\theta,v}(x,t) = N^2 \int b(N(\xi - \nu_\theta))e^{2\pi i((x-v)\cdot\xi + t\phi(\xi))}\,d\xi. \tag{4.1}$$

The labels ν_θ form a separated net as in Section 2. For each θ, v ranges over a fixed rectangular lattice with spacing comparable to N, with an arbitrary translate. A fixed finite collection of profiles is allowed. All their smoothness bounds and lattice constants are fixed before the loss parameters vary. Put

$$D_a(x,t) = \sum_{\theta,v} |a_{\theta,v}|^2 \langle N^{-1}(x - v + tV_\theta)\rangle^{-M}, \qquad V_\theta = \nabla\phi(\nu_\theta). \tag{4.2}$$

Set, with a fixed multiplicative constant enlarged whenever necessary,

$$\mathcal{A}(R) = P(\kappa,\eta)^{1/3} R^{(10\kappa+\eta)/6} \log^{2/3}(2 + R). \tag{4.3}$$

The constants implicit below do not depend on κ, η. A single choice of the fixed label bounds in P suffices for the finitely many constructions used in the proof.

Proposition 2. *External input 2 implies*

$$\left\| \sum_{\theta,v} a_{\theta,v} P^b_{\theta,v} \right\|_{L^3(\mathbb{R}^2\times[0,R])} \le C\mathcal{A}(R) \left\| D_a^{1/2} \right\|_{L^3(\mathbb{R}^2\times[0,R])} \tag{4.4}$$

for every finite complex array, with no restriction on the locations of its spatial labels. The same estimate holds after translating the common time origin.

We prove the proposition in three steps, including the localization that removes the position restriction in the external input.

4.1. **Continuous reconstruction in a bounded region.** First use exactly the windows in (3.3), with $|v| \le C_1 R$. Fix a square Q of side comparable to R in a fixed multiple of $[-R, R]^2$. For $j = 0, \dots, R - 1$, Fourier series and (3.2) give the exact identity

$$g_d(\xi)e^{2\pi i(j\phi(\xi))} = D^{-2} \sum_{\ell,w} c_{\ell,w,j}\chi^1_\ell(\xi)e^{2\pi i(-w\cdot\xi)}. \tag{4.5}$$

Indeed, for each ℓ, take the Fourier series of $g_d\chi^1_\ell e^{2\pi i(j\phi)}$ on the square of side D containing its support, and then multiply by the second copy of χ^1_ℓ. Summing in ℓ uses (3.2). The

factor D^{-2} is the reciprocal area of that square. Constant phases caused by translating the square cancel in the analysis and synthesis.

For $t = j + u$, $0 \le u \le 1$, this gives

$$F_d(x,t) = D^{-2} \sum_{\ell,w} c_{\ell,w,j} K_\ell(x - w, u), \quad K_\ell(z,u) = \int \chi_\ell^1(\xi) e^{2\pi i (z\cdot\xi + u\phi(\xi))}\, d\xi.$$

Integration by parts in ξ shows, for every fixed L,

$$|K_\ell(z,u)| \le C_L \langle z \rangle^{-L} \qquad (0 \le u \le 1). \tag{4.6}$$

There are only a fixed number of ℓ. The sums of these bounds over $w \in D^{-1}\mathbb{Z}^2$, and their integrals in z, are bounded. Weighted Hölder therefore bounds the integral of the cube of the reconstruction using $|w| \le C_2 R$ by

$$C \sum_{\ell,\, |w| \le C_2 R,\, j} |c_{\ell,w,j}|^3. \tag{4.7}$$

For example, $(\sum |c_i| k_i)^3 \le (\sum k_i)^2 \sum |c_i|^3 k_i$; integrating (4.6) proves the assertion.

Here C_2 is chosen larger than the fixed position bound for Q. For the omitted terms, $|c_{\ell,w,j}| \le C \sum |d|$ directly from (3.3), since each frequency window has area $O(N^{-2})$. For $x \in Q$,

$$\sum_{|w| > C_2 R} \langle x - w \rangle^{-L} \lesssim R^{2-L}.$$

There are at most CN^4 initial indices: $O(N^2)$ frequency labels and $O(N^2)$ position labels. Consequently the omitted reconstruction has cubed $L^3(Q \times [0,R])$ norm at most

$$C_L R^3 R^{6-3L} \Big(\sum |d| \Big)^3 \le C_L N^{26-6L} \sum |d|^3.$$

On the other hand, superadditivity of $s^{3/2}$ on nonnegative numbers gives

$$\int_0^R \int (H_d^0)^{3/2}\, dx\, dt \ge \sum_{\nu,v} |d_{\nu,v}|^3\, \pi N^2 R = \pi N^4 \sum |d|^3. \tag{4.8}$$

Choose $L \ge 5$. The omitted terms are controlled by (4.8). Combining this with (4.7) and the external input proves

$$\|F_d\|_{L^3(Q\times[0,R])} \le C\mathcal{A}(R) \left(\int_0^R \int (H_d^0)^{3/2} \right)^{1/3}. \tag{4.9}$$

Thus continuous reconstruction has introduced no power of R.

4.2. **Changing packet windows and lattices.** Suppose now that $|v| \le C_3 R$ in an arbitrary packet system (4.1). Its initial frequency datum is

$$g(\xi) = N^2 \sum_{\theta,v} a_{\theta,v}\, b(N(\xi - \nu_\theta)) e^{2\pi i(-v\cdot\xi)}.$$

Expand this datum in the exact fine frame with windows χ_μ^δ and labels $w \in (N/D)\mathbb{Z}^2$:

$$d_{\mu,w} = \int g(\xi) \chi_\mu^\delta(\xi) e^{2\pi i (w\cdot\xi)}\, d\xi, \qquad g = (D\delta)^{-2} \sum_{\mu,w} d_{\mu,w} \chi_\mu^\delta e^{2\pi i(-w\cdot\xi)}. \tag{4.10}$$

The verification is the same Fourier-series argument as for (4.5). There is a matrix B with $d_{\mu,w} = \sum B_{\mu,w;\theta,v} a_{\theta,v}$, and

$$|B_{\mu,w;\theta,v}| \le C_L \langle (w - v)/N \rangle^{-L}, \qquad B_{\mu,w;\theta,v} = 0 \text{ if } |\mu - \nu_\theta| > C/N. \tag{4.11}$$

To prove it, substitute $\xi = \nu_\theta + \zeta/N$ in the coefficient integral and integrate by parts in ζ. The normalized amplitudes have uniformly bounded derivatives. Bounded overlap gives only a bounded number of interacting frequency labels.

Weighted Cauchy–Schwarz and (2.2) imply, for $0 \le t \le R$,

$$\sum_{\mu,w} |d_{\mu,w}|^2 \mathbf{1}_{\{|x-w+t\nabla\phi(\mu)|\le N\}} \le C D_a(x,t). \tag{4.12}$$

In detail, first bound $|d_{\mu,w}|^2$ by a constant times the sum of $|a_{\theta,v}|^2\langle (w-v)/N\rangle^{-L}$. On the displayed core, convolution with this weight gives the weight in (4.2). Replacing $\nabla\phi(\mu)$ by V_θ changes the centre by at most $CR/N = CN$, which is one fixed multiple of the transverse width.

The new array is not finitely supported in w. Retain $|w| \le C_4 R$, with C_4 sufficiently larger than C_3. From (4.11), summing the omitted positions gives

$$\sum_{\mu,\,|w|>C_4R} |d_{\mu,w}| \le C_L N^{2-L} \sum_{\theta,v} |a_{\theta,v}|.$$

Each synthesized free packet has size bounded by a constant, for all x, t, directly from its frequency integral. Hence on $Q \times [0,R]$ the omitted synthesis has cubed norm at most

$$C_L N^6 N^{6-3L} \Big(\sum |a| \Big)^3 \le C_L N^{20-3L} \sum |a|^3.$$

The original bounded-position array has at most CN^4 indices. Also,

$$\int_0^R \int D_a^{3/2}\, dx\, dt \ge c_M N^4 \sum |a|^3, \tag{4.13}$$

by integrating each translated weight and using superadditivity. Taking $L \ge 6$, and larger if needed for (4.12), controls the omitted synthesis. Apply (4.9) to the retained array and use (4.12). We have proved

$$\Big\| \sum a_{\theta,v} P^b_{\theta,v} \Big\|_{L^3(Q\times[0,R])} \le C\mathcal{A}(R) \left(\int_0^R \int D_a^{3/2} \right)^{1/3} \tag{4.14}$$

for every bounded-position array. All position bounds used here are fixed constants times R.

4.3. **Removing the position restriction.** Tile $\mathbb{R}^2$ by the squares $Q_m = mR + [-R/2, R/2)^2$, $m \in \mathbb{Z}^2$. On $Q_m \times [0,R]$ split the array into $|v - mR| \le C_5 R$ and its complement, with C_5 larger than a fixed multiple of $1 + \sup|V_\theta|$. Translation by mR permits (4.14) for the first part. The canonical lattices are preserved since D, N are integers. The original lattices may translate, which was already allowed.

Let D_m be the density of this first part. Each initial label belongs to at most a fixed number of such parts. Pointwise,

$$\sum_m D_m^{3/2} \le \Big(\sum_m D_m \Big)^{3/2} \le C D_a^{3/2}.$$

Thus the cubes of the near-part estimates sum to at most $C\mathcal{A}(R)^3 \int D_a^{3/2}$.

For the far part, Taylor expansion in the packet integral gives, for every L,

$$|P^b_{\theta,v}(x,t)| \le C_L \langle (x - v + tV_\theta)/N\rangle^{-L}, \qquad 0 \le t \le R. \tag{4.15}$$

Indeed the quadratic remainder has uniformly bounded derivatives after normalization, since $t/N^2 \le 1$. On the far part its normalized spatial argument has size at least cN. For each fixed frequency the sum over the spatial lattice of its power $-2L + M$ is at most CN^{M+2-2L}. There are $O(N^2)$ frequencies. Cauchy–Schwarz therefore gives, on $Q_m \times [0,R]$,

$$\left| \sum_{|v-mR|>C_5R} a_{\theta,v} P^b_{\theta,v} \right|^2 \le D_a(x,t)\, CN^{M+4-2L} \le C D_a(x,t)$$

once $2L > M + 4$. Integrating over the disjoint squares finishes the proof of Proposition 2. Translation of the time coordinate proves its last assertion.

For any prescribed $\alpha > 0$, (4.3) is at most $C_\alpha R^\alpha$ after choosing fixed small κ, η. More usefully, we have preserved its explicit dependence throughout the proof.

5. Changes of time origin

Recall that $r = R^{1/2} = N$. Put

$$A_\theta(x,t) = \Big(r^{-1}(x + tV_\theta), R^{-1}t\Big)$$

and define the weight centred at (v,s) by

$$W_{\theta,v,s}(x,t) = \langle r^{-1}(x - v + (t-s)V_\theta)\rangle^{-M}\langle R^{-1}(t-s)\rangle^{-M}. \tag{5.1}$$

Its integral is comparable to $r^2R = R^2$, since the change of variables A_θ has determinant R^{-2}.

Fix a Schwartz function ψ on $\mathbb{R}$, and set

$$p_{\theta,v,k}(x,t) = \psi\Big((t-kR)/R\Big)P^b_{\theta,v}(x, t-kR), \qquad k \in \mathbb{Z}. \tag{5.2}$$

For a finite coefficient array a, let

$$H_a = \sum_{\theta,v,k} |a_{\theta,v,k}|^2 W_{\theta,v,kR}. \tag{5.3}$$

Proposition 3. *For $\mathcal{A}(R)$ defined in (4.3),*

$$\left\| \sum_{\theta,v,k} a_{\theta,v,k} p_{\theta,v,k} \right\|_{L^3(\mathbb{R}^3)} \le C\mathcal{A}(R) \left\| H_a^{1/2} \right\|_{L^3(\mathbb{R}^3)}. \tag{5.4}$$

The conclusion also holds for any fixed oversampling of the time lattice and for fixed translates of the spatial lattices depending on θ and k.

Proof. Let $I_j = [jR, (j+1)R]$. Fix k, and denote its free wave by

$$F_k(x,t) = \sum_{\theta,v} a_{\theta,v,k} P^b_{\theta,v}(x, t-kR).$$

We re-expand F_k at time jR. We give the coefficient estimate, since it is the point that permits use of (4.4) with constants independent of the time origins.

Use the fine frame (4.10), with $\vartheta \in r^{-1}\mathbb{Z}^2$ and $w \in (r/D)\mathbb{Z}^2$. In this paragraph its frequency datum and synthesis coefficients are explicitly

$$g_k(\xi) = r^2 \sum_{\theta,v} a_{\theta,v,k} b(r(\xi - \nu_\theta)) e^{2\pi i(-v\cdot\xi)}, \qquad c^{j,k}_{\vartheta,w} = D^{-2} \int g_k \chi_\vartheta^{1/r} e^{2\pi i(w\cdot\xi + (j-k)R\phi(\xi))}\, d\xi.$$

Here the fixed integer D is the side length chosen in Section 3, and $V_\vartheta = \nabla\phi(\vartheta)$. The frame identity gives

$$F_k(x,t) = \sum_{\vartheta,w} c^{j,k}_{\vartheta,w} P^\chi_{\vartheta,w}(x, t - jR). \tag{5.5}$$

Only a bounded number of ϑ interact with each θ, and conversely, because their frequency windows must overlap.

Put $\Delta = j - k$. The matrix entry carrying $a_{\theta,v,k}$ to $c^{j,k}_{\vartheta,w}$ is, up to a fixed normalization and a factor of modulus one,

$$K_{\vartheta,\theta,\Delta}(u) = \int B_{\vartheta,\theta}(\zeta) e^{2\pi i(u\cdot\zeta + \Delta q_\theta(\zeta))}\, d\zeta, \qquad u = \frac{w - v + \Delta R V_\theta}{r}, \tag{5.6}$$

where $B_{\vartheta,\theta}(\zeta) = b(\zeta)\chi(\zeta + r(\nu_\theta - \vartheta))$ has uniformly bounded derivatives and fixed bounded support, and

$$q_\theta(\zeta) = R\left[\phi(\nu_\theta + r^{-1}\zeta) - \phi(\nu_\theta) - r^{-1}V_\theta\cdot\zeta\right]. \tag{5.7}$$

Taylor's theorem gives a uniform bound for every fixed derivative of q_θ on this support. For derivatives of order $m \geq 2$, the scale factor is $Rr^{-m} = R^{1-m/2} \leq 1$.

Integrating by parts in (5.6) gives, for every L,

$$|K_{\vartheta,\theta,\Delta}(u)| \leq C_L(1+|\Delta|)^{C_L}\langle u\rangle^{-L}. \tag{5.8}$$

This estimate is useful even for large $|\Delta|$: the loss is polynomial in $|\Delta|$, and the time cutoff in (5.2) decreases faster than any power.

Choose L larger than $M+10$. Weighted Cauchy–Schwarz, the bounded frequency overlap, and (2.2) imply

$$\begin{aligned}&\sum_{\vartheta,w} |c^{j,k}_{\vartheta,w}|^2 \langle r^{-1}(x-w+(t-jR)V_\vartheta)\rangle^{-M} \\ &\qquad \leq C(1+|j-k|)^C \sum_{\theta,v} |a_{\theta,v,k}|^2 \langle r^{-1}(x-v+(t-kR)V_\theta)\rangle^{-M}, \qquad t\in I_j.\end{aligned} \tag{5.9}$$

Indeed, convolution of the two spatial weights centres the result at $x-v+(t-kR)V_\theta$. The replacement of V_ϑ by V_θ changes the normalized centre by $O(1)$, since $|V_\vartheta - V_\theta| \lesssim r^{-1}$ and $|t-jR| \leq R$.

Write the sum on the right of (5.9), without its constant, as $H_k(x,t)$. Applying (4.4) to (5.5) yields

$$\|F_k\|_{L^3(\mathbb{R}^2\times I_j)} \leq C\mathcal{A}(R)(1+|j-k|)^C \left\|H_k^{1/2}\right\|_{L^3(\mathbb{R}^2\times I_j)}. \tag{5.10}$$

Infinite spatial coefficient arrays arising in (5.5) are handled by finite truncation. The decay in (5.8) gives convergence of the coefficient densities and of synthesis on each slab, so (4.4) passes to this limit.

For any $A > 1$, the Schwartz bound for ψ absorbs the polynomial factor in (5.10). Minkowski's inequality therefore gives

$$\left\|\sum_k \psi((t-kR)/R)F_k\right\|_{L^3(\mathbb{R}^2\times I_j)} \leq C_A\mathcal{A}(R)\sum_k (1+|j-k|)^{-A}\left\|H_k^{1/2}\right\|_{L^3(\mathbb{R}^2\times I_j)}.$$

For $q_k \geq 0$ and $u_k \geq 0$, Hölder's inequality gives $(\sum q_k u_k)^3 \leq (\sum q_k)^2 \sum q_k u_k^3$. Use this with $q_k = (1+|j-k|)^{-A}$, where $A > 1$, to obtain

$$\begin{aligned}&\left\|\sum_k \psi((t-kR)/R)F_k\right\|^3_{L^3(\mathbb{R}^2\times I_j)} \\ &\qquad \leq C\mathcal{A}(R)^3 \int_{\mathbb{R}^2\times I_j} \sum_k (1+|j-k|)^{-A} H_k^{3/2} \\ &\qquad \leq C\mathcal{A}(R)^3 \int_{\mathbb{R}^2\times I_j} \left(\sum_k (1+|j-k|)^{-2A/3} H_k\right)^{3/2}.\end{aligned} \tag{5.11}$$

Choose $2A/3 \geq M$. On I_j, the weight in this last sum is bounded by a constant times $\langle (t-kR)/R\rangle^{-M}$. Summing (5.11) in j proves (5.4).

A fixed oversampled time lattice can be chosen to have spacing R/B, where B is a fixed integer. Split its indices into the B residue classes. Each class is a translate of the time lattice above. A final application of Hölder's inequality costs only a constant depending on B. The spatial arguments use (2.2), which is uniform under translations of the lattices. □

6. An expansion for functions in a thickened surface

We next construct the coefficients to which Proposition 3 will be applied. This step is necessary because the functions in Theorem 1 need not be free extension waves.

Lemma 4. *For every family f_θ in Theorem 1, there is a packet expansion*

$$f_\theta = \sum_{v,k} a_{\theta,v,k} p_{\theta,v,k}, \tag{6.1}$$

with fixed profiles and fixed oversampling constants as in Proposition 3. Its density H_a satisfies

$$H_a(y) \le C \sum_\theta (|f_\theta|^2 * K_\theta)(y), \tag{6.2}$$

where

$$K_\theta(x,t) = R^{-2} \langle r^{-1}(x + tV_\theta)\rangle^{-M} \langle R^{-1}t\rangle^{-M}. \tag{6.3}$$

The constants are independent of R and of the family f_θ.

Proof. Choose smooth functions χ, ρ with compact support, with $\chi = 1$ on the normalized angular supports and $\rho = 1$ on $[-C_0, C_0]$. Then

$$m_\theta(\xi,\lambda) = \chi(r(\xi - \nu_\theta))\rho(R(\lambda - \phi(\xi))) \tag{6.4}$$

equals one on $\operatorname{supp} \widehat{f_\theta}$.

Introduce the affine frequency coordinates

$$\eta = r(\xi - \nu_\theta), \qquad \sigma = R\Big(\lambda - \phi(\nu_\theta) - V_\theta \cdot (\xi - \nu_\theta)\Big). \tag{6.5}$$

Taylor's theorem implies that the supports of both m_θ and $\widehat{f_\theta}$ lie in a fixed rectangular box in (η, σ). Choose a slightly larger box Q whose spatial side length is a fixed number A and whose last side length is a fixed integer B. Take all supports in its interior. The corresponding frequency box Q_θ has volume

$$|Q_\theta| = A^2 B r^{-2} R^{-1} = A^2 B R^{-2}.$$

Fourier series on Q_θ give

$$\widehat{f_\theta}(\zeta) = \sum_{\ell \in \mathbb{Z}^2,\, k \in \mathbb{Z}} c_{\theta,\ell,k} e^{2\pi i\left(-y_{\theta,\ell,k}\cdot\zeta\right)} \quad \text{on } Q_\theta, \tag{6.6}$$

after absorbing constant phases into the coefficients, where

$$s_k = kR/B, \qquad v_{\theta,\ell,k} = r\ell/A - s_k V_\theta, \qquad y_{\theta,\ell,k} = (v_{\theta,\ell,k}, s_k). \tag{6.7}$$

In particular,

$$|c_{\theta,\ell,k}| = |Q_\theta|^{-1} |f_\theta(y_{\theta,\ell,k})|. \tag{6.8}$$

Multiply (6.6) by m_θ, and take its inverse Fourier transform. Direct integration in λ yields

$$\begin{aligned} \check{m}_\theta(x,t) &= R^{-1}\check{\rho}(t/R) \int \chi(r(\xi - \nu_\theta)) e^{2\pi i(x\cdot\xi + t\phi(\xi))}\, d\xi \\ &= R^{-2}\check{\rho}(t/R) P^\chi_{\theta,0}(x,t). \end{aligned} \tag{6.9}$$

Thus (6.1) holds with $\psi = \check{\rho}$, time lattice $R\mathbb{Z}/B$, spatial centres in (6.7), and

$$a_{\theta,\ell,k} = R^{-2} c_{\theta,\ell,k}, \qquad |a_{\theta,\ell,k}| = (A^2 B)^{-1} |f_\theta(y_{\theta,\ell,k})|. \tag{6.10}$$

The affine coordinates in (6.5) are used only for the Fourier series. The multiplier in (6.4) retains the exact phase ϕ, which is why (6.9) gives free elliptic packets rather than packets for a tangent plane.

We now estimate the samples. Choose a smooth cutoff on a slightly enlarged tangent box that equals one on $\operatorname{supp} \widehat{f_\theta}$, and denote its inverse Fourier transform by κ_θ. Then $f_\theta = f_\theta * \kappa_\theta$, and, for every L,

$$|\kappa_\theta(y)| \le C_L R^{-2} \langle r^{-1}(x + tV_\theta)\rangle^{-L} \langle R^{-1}t\rangle^{-L}, \qquad \|\kappa_\theta\|_{L^1} \le C. \tag{6.11}$$

Weighted Cauchy–Schwarz therefore gives

$$|a_{\theta,\ell,k}|^2 \le C_L R^{-2} \int |f_\theta(z)|^2 \langle r^{-1}(z_x - v_{\theta,\ell,k} + (z_t - s_k)V_\theta)\rangle^{-L} \langle R^{-1}(z_t - s_k)\rangle^{-L}\, dz. \tag{6.12}$$

Multiply (6.12) by $W_{\theta,v_{\theta,\ell,k},s_k}(y)$ and sum in ℓ, k. Under A_θ, the centres in (6.7) become exactly the fixed lattice $(\ell/A, k/B)$. The product version of (2.2), with L sufficiently large compared with M, bounds this sum by $CK_\theta(y-z)$. Integrating in z and then summing in θ proves (6.2).

All equalities can first be read in L^2 by Fourier-series convergence. Since $\widehat{f_\theta}$ is smooth and supported in the interior of Q_θ, its coefficients decrease faster than every power at each fixed R. The expansions consequently converge in the norms used below. Equivalently, one can truncate the series and pass to the limit using (6.2) and Proposition 3. □

7. Tube averages and the proof of the theorem

The following positive estimate is the part of the proof that uses the Nikodym hypothesis.

Lemma 5. *Assume External input 3, and let K_θ be as in* (6.3)*. For every $0 < \gamma \le 1$ and every finite family of nonnegative functions $g_\theta \in L^{3/2}(\mathbb{R}^3)$,*

$$\left\| \sum_\theta g_\theta * K_\theta \right\|_{L^{3/2}} \le CM(\gamma) R^{\gamma/2} \left\| \sum_\theta g_\theta \right\|_{L^{3/2}}. \tag{7.1}$$

Proof. Set $G = \sum_\theta g_\theta$, and write $\widetilde{K_\theta}(y) = K_\theta(-y)$. For $h \ge 0$, Tonelli's theorem gives

$$\begin{aligned} \int \left(\sum_\theta g_\theta * K_\theta \right) h &= \sum_\theta \int g_\theta (\widetilde{K_\theta} * h) \\ &\le \int G \sup_\theta (\widetilde{K_\theta} * h). \end{aligned} \tag{7.2}$$

The set $T_\theta = \{y : |r^{-1}(x + tV_\theta)| \le 1,\ |t/R| \le 1\}$ is a sheared box of volume comparable to R^2. Since V_θ is bounded, it is contained in a circular tube of length comparable to R and radius comparable to r, and the reverse containment holds after fixed enlargements.

The product weight in (6.3) is bounded by $CR^{-2}(1 + |A_\theta y|)^{-M}$. Decomposing into dyadic annuli in these normalized coordinates gives

$$K_\theta(y) \le C \sum_{m=0}^{\infty} 2^{-m(M-3)} \frac{\mathbf{1}_{C2^m T_\theta}(y)}{|C2^m T_\theta|}. \tag{7.3}$$

Take $M > 4$. The tubes in the mth term have length $L_m \asymp 2^m R$ and eccentricity comparable to $R^{-1/2}$. Thus

$$\sup_\theta (\widetilde{K_\theta} * h) \le C \sum_{m=0}^{\infty} 2^{-m(M-3)} \mathcal{N}_{c_1 R^{-1/2}, L_m} h,$$

with a fixed dimensional constant c_1. For bounded R the estimate follows directly from fixed-eccentricity averaging; otherwise the Nikodym parameter is less than one. Apply (3.8) with exponent γ, use its dilation invariance, and sum the geometric series. We obtain

$$\left\| \sup_\theta (\widetilde{K_\theta} * h) \right\|_{L^3} \le CM(\gamma) R^{\gamma/2} \|h\|_{L^3}. \tag{7.4}$$

Hölder's inequality in (7.2) now gives

$$\int \left(\sum_\theta g_\theta * K_\theta \right) h \le CM(\gamma) R^{\gamma/2} \|G\|_{L^{3/2}} \|h\|_{L^3}.$$

Duality proves (7.1). In particular, we have not interchanged a norm with an uncontrolled supremum over tube lengths: the lengths were summed with decreasing coefficients. □

Proof of Theorem 1 for a continued graph. Use Lemma 4 to expand the cap functions. Proposition 3 and the coefficient-density bound give

$$\|f\|_{L^3} \le C\mathcal{A}(R)\left\|H_a^{1/2}\right\|_{L^3}, \qquad \|H_a\|_{L^{3/2}} \le CM(\gamma)R^{\gamma/2}\left\|\sum_\theta |f_\theta|^2\right\|_{L^{3/2}}.$$

Taking the square root of the second inequality proves the quantitative bound

$$\left\|\sum_\theta f_\theta\right\|_{L^3} \le CP(\kappa,\eta)^{1/3}M(\gamma)^{1/2}R^{(10\kappa+\eta)/6+\gamma/4}\log^{2/3}(2+R)\,\|\mathcal{S}(\mathbf{f})\|_{L^3}. \tag{7.5}$$

Choose fixed positive κ, η, γ so that the exponent of R displayed here is less than $\varepsilon/2$. The remaining logarithm is at most $C_\varepsilon R^{\varepsilon/2}$.

So far $R = N^2$ with N a power of two. For arbitrary $R \ge 2$ choose such an N with $R \le N^2 < 4R$. The original caps have the admissible dimensions at scale N^2 after a fixed enlargement, and retain uniform separation. Apply the result at that scale. The fixed factors are absorbed into the constants. The passage to other graph phases and compact surfaces is proved next. □

8. General graphs and compact surfaces

For a smooth embedded surface, the second fundamental form is the quadratic form measuring its normal curvature on each tangent plane. Definiteness means that its two eigenvalues have the same nonzero sign. Choosing the sign of the normal makes it positive definite locally. In tangent coordinates the surface is then a graph with positive definite Hessian. On a sufficiently small compact chart the Hessian satisfies (1.1).

Let Σ be a compact such surface, possibly with smooth boundary. At a boundary point we use a smooth extension of the local parametrization across the boundary. The inverse function theorem gives an extended graph there, and continuity preserves positive definiteness after shrinking the chart. No estimate is required for a nonsmooth extension by zero of the phase.

Choose a maximal δ-separated set of points p on Σ. Define its thickened caps by

$$\theta_p = \{\zeta \in \mathbb{R}^3 : \operatorname{dist}(\zeta, \Sigma \cap B(p, C_0\delta)) \le C_0\delta^2\}. \tag{8.1}$$

Here $\operatorname{dist}(\zeta, E) = \inf_{q\in E}|\zeta - q|$. Increasing C_0 by a fixed factor ensures that these caps cover the δ^2-neighbourhood of Σ. Each fixed chart is bi-Lipschitz on its compact domain, so the disk-packing argument of Section 2 gives bounded overlap and $O(\delta^{-2})$ caps. Taylor's formula in that chart puts every cap in a box of dimensions δ, δ, δ^2. At boundary points we only need containment in such a box, not a full box inside the cap. Smooth partitions are obtained by the bump-and-denominator construction of Section 2 in finitely many charts.

Cover Σ by finitely many small graph charts, and then use External input 1 on smaller charts. Choose fixed ambient smooth cutoffs $q_j(\zeta)$ subordinate to these charts, with

$$\sum_{j=1}^{J} q_j(\zeta) = 1$$

on a fixed neighbourhood of Σ. The number J and the cutoffs are independent of R. Let Q_j be the Fourier multiplier with symbol q_j, and let $k_j = \check{q}_j$; thus $Q_j f = k_j * f$ and $\|k_j\|_1 < \infty$.

For all sufficiently large R, every cap meeting $\operatorname{supp} q_j$ has its centre in a slightly enlarged version of the same chart. The graph coordinate map and the fixed affine placement carry these centres to a separated planar net, with changed fixed separation constants. Taylor's formula carries their caps into fixed enlargements of (1.2). Only a bounded number of neighbours can coalesce at the scale δ; if necessary divide them into finitely many separated families. One can do this by greedily assigning colours in a graph in which nearby centres

are joined: the packing bound gives a fixed bound on the number of neighbours, hence on the colours. Summing the resulting finitely many inequalities costs a fixed constant.

For completeness, an affine change of frequency coordinates $\zeta \mapsto A\zeta + b$, with A invertible, corresponds to

$$g(y) = e^{2\pi i(b\cdot y)} f(A^T y), \qquad \widehat{g}(\eta) = |\det A|^{-1} \widehat{f}(A^{-1}(\eta - b)).$$

Both sides of the square function inequality are multiplied by $|\det A|^{-1/3}$. Frequency translations only insert the common factor of modulus one. Thus these coordinate changes preserve the desired inequality, with the stated geometric changes of constants.

Apply the continued-graph estimate to the family $(Q_j f_\theta)_\theta$ after its coordinate change. Its square function is bounded by that of the original family, since Minkowski's inequality in the Euclidean norm of a finite sequence gives pointwise

$$\left(\sum_\theta |Q_j f_\theta(y)|^2\right)^{1/2} \le \int |k_j(z)| \left(\sum_\theta |f_\theta(y-z)|^2\right)^{1/2} dz.$$

Young's convolution inequality therefore gives

$$\left\| \left(\sum_\theta |Q_j f_\theta|^2\right)^{1/2} \right\|_{L^3} \le \|k_j\|_1 \, \|\mathcal{S}(\mathbf{f})\|_{L^3}.$$

Sum over the fixed collection of charts. The bounded range of remaining R is covered by the pointwise estimate $|\sum f_\theta| \le (\#\{\theta\})^{1/2} \mathcal{S}(\mathbf{f})$ and the bounded number of caps in that range. This proves the last assertion of Theorem 1. The same argument, applied to a compact graph, removes the continued-phase restriction in its first assertion.

This argument also describes the partition used for a closed surface such as the sphere: first choose finitely many fixed graph cutoffs, then choose the δ-scale partition in each graph. All partitions are smooth frequency partitions. We never multiply a function in physical space by the indicator of a tube to define its Fourier pieces.

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